Daniel Gonzalez Lazaro
Team 2
Yechen Zheng, Hongfei Liu, Hongru Liu
ILR06
3/17/2026

Team progress

The team has moved closer to a final design that will be constructed by specifying the materials, bearings, and ideating manufacturing processes to actually produce and drive the linked legs while supporting the mechanical loads. We have therefore ordered the parts to fully construct one of these legs, which should quite directly translate to the other six given that all of them share the same design. Additionally parts needed to finish the aluminum chassis ‘cage’ have been ordered, and the cad updated to show mounting of motors, gearboxes, drivers, etc. We will continue the assembly as soon as parts arrive.

Individual progress

I ended up being responsible with selecting/estimating what motors, drivers, gearboxes would be necessary for the robot after having budget difficulties with selecting all in one units from myactuator or cubemars, given these are sold from China with high duties, not included in the cost used to estimate budget. My selected servomotor / driver combos ended up being stepperonline RS485-driven units. These come out to a total cost around half of the motor stipend, since I realized that the real main constraint for motor selection would be weight, not cost, of both the motor units and the corresponding battery of a large enough size to supply current with a reasonable expectation of lasting 30 minutes in continuous motion, as expected by the design requirements. Given we are also specifically constrained to 150 lbs, this is much more restrictive than cost given CAD assembly estimations excluding these components is 70-80 lbs. To get more information for informed decisions, I made substantial progress in our MuJoCo simulation environment, adding a humanoid (weighing 180lb) load on top of the robot and for these purposes, sensing and .csv logging of torque over time, as well as having torque constraints reflecting our motor candidates. By logging torque over time, and using the specification of 0.127 Nm / A provided by the manufacturer, it is probable that a 30 min runtime would consume less than 30 ah, so acceptable batteries that are lightweight enough would be in the 25-30 ah range, mostly restricted to lithium ion. The reason that the servos are RS485 specifically, is since it was this version of the motor that was available for shipping from U.S. warehouses. Thus I had to change my design to use the appropriate TTL module as opposed to a CAN to TTL. Since the drivers actually utilize RJ45 connectors to carry the RS485 signal (though only using 3 of the 8 data lines supported by the connector), this also included adding a ethernet breakout and connecting it to that module.

I’ve also been able to update the control to an FSM of gait states. The main idea is to package two CPGs, one ripple and one tripod, and move between them with a tuneable threshold of forward and yaw commanded rates. The ripple, with proper contact sensing should allow for more stable movement since it entails having each shifted by less phase than tripod (pi / 6), so more frequent contact should be possible. This helps especially in low speed turning or turning in place since the specific leg design trade off of using a linkage to drive motion with one motor instead of multiple does introduce by its nature kinematic constraints that an insect tripod gait would avoid. Ive also set up a mujoco MJX training environment for this new version, rolling out generations of identical robots to follow the movement specifications of arc motion, forward and backward with a variable angle range of 0-90 degrees, where the session allows a somewhat automatic testing of the gains of the different controller parameters on a reward function of the least effort, stable following of the established arc. In theory this should aid in minimizing power consumption in driving useless motion.

Finally, I’ve gone ahead and set up the microcontroller circuit, IMU, LCDs, and all other components excluding the FSRs on a soldered pin board, which took an unexpectedly long time. For modularity and how these components would actually be mounted in different locations within the robot, I made three separate boards, two adapters for the used pins of the LCDs, and one main

board including the esp, IMU, and pin header connectors for the TTL module and FSRs. The FSRs are three pins, 3V3, ADC, and GND, and the ttl is 3V3, TX, RX, GND. Thus those components that need to be mounted (close to drivers, or on a visible face of the robot) are separate and connected with the minimum amount of F/F DuPont connectors.

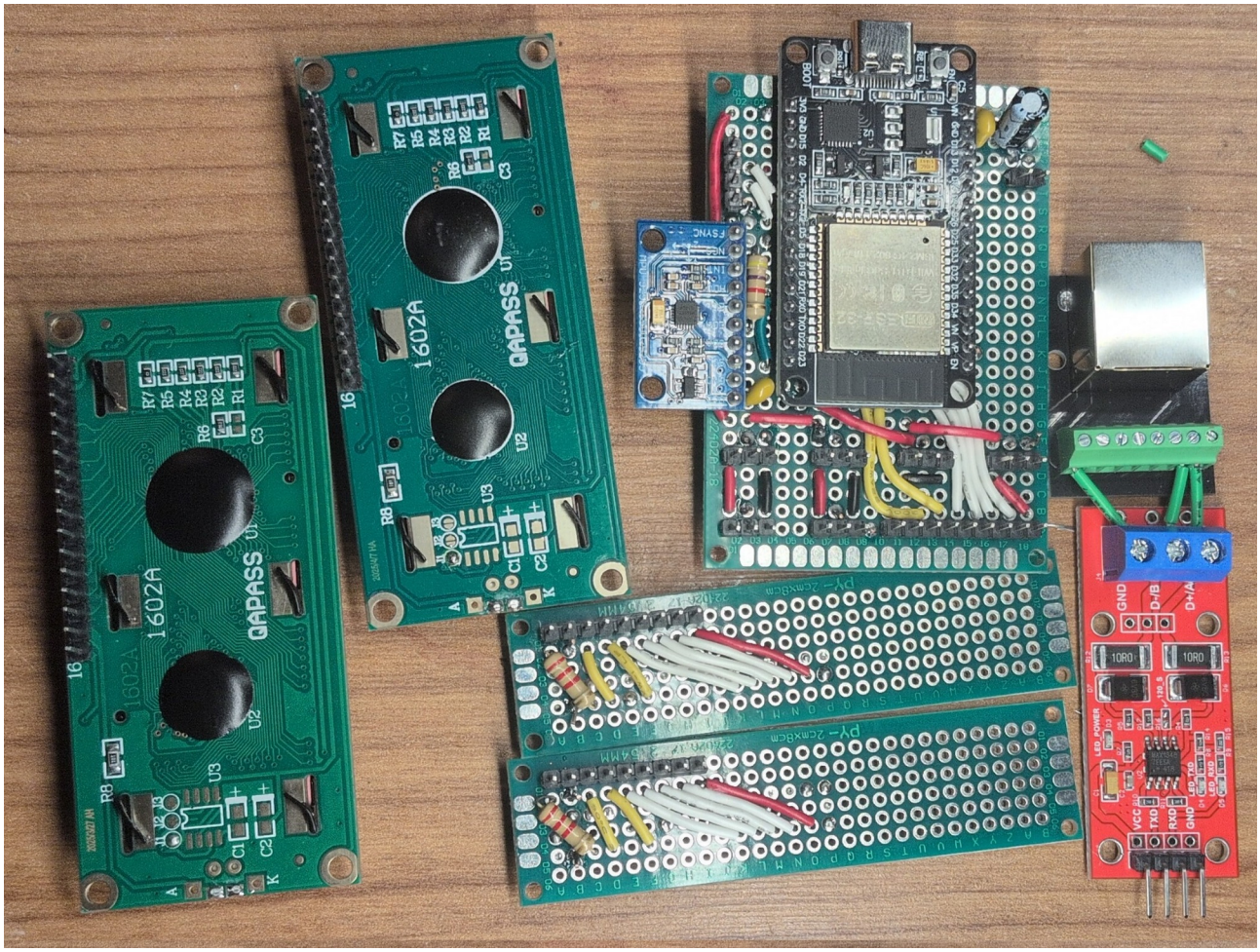

Figure 1: Pinboard and related components

Next Steps

- Drafting up and flashing RS485 drivers to communicate with the servos so that I can test out the servos as soon as possible once they arrive.
- Carefully test connections on the pin boards to make sure that there are no shorts or unexpected connections before powering on to avoid breaking components.
- Finalize schematic / part list as it relates to the full power system, focusing on safety given the large lithium ion battery and substantial currents needed to drive the servomotors. My objective is to order the battery tomorrow, and ideally the other components to safely connect to the motor / compute as well.
- Begin implementing nav2 and slam on the simulation to be the as similar as possible as the jetson to test out the most comprehensive functional architecture as possible on the simulation to verify function.
- Mount compute on the chassis and test out basic IMU / camera physical SLAM.